

Diabetic Retinopathy Detect

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Chapter 1

Introduction

Diabetic Retinopathy (DR) is a leading cause of preventable blindness in diabetic patients. Early detection and classification of DR stages are critical to prevent vision loss. With advancements in deep learning, automated systems for DR detection using retinal fundus images have shown promise in enhancing the efficiency and accuracy of diagnosis.

1.1 Problem Domain

Traditional methods for diagnosing Diabetic Retinopathy rely on manual examination of retinal images by specialists, which can be subjective and time-consuming. Convolutional Neural Networks (CNNs) have shown effectiveness in automating this process by extracting local features from retinal images. However, CNNs are limited in capturing global dependencies in the image, which are essential for understanding complex retinal structures. On the other hand, Vision Transformers (ViTs) excel in capturing long-range dependencies but require large datasets and computational resources, making them less practical for medical imaging tasks with limited data availability.

1.2 Research Problem Statement

The challenge is to develop an efficient and accurate system for Diabetic Retinopathy detection that leverages the strengths of both CNNs and ViTs. Specifically, how can a hybrid CNN-ViT model be designed to enhance feature extraction and classification accuracy for DR detection, while addressing the limitations of each individual architecture? This research aims to investigate the viability of such a hybrid model and its effectiveness in improving Diabetic Retinopathy detection.

Chapter 2

Literature Review

2.1 Related Research

2.1.1 Is it Time to Replace CNNs with Transformers for Medical Images?

2.1.1.1 Summary

The paper[1] investigates whether Vision Transformers (ViTs) can effectively replace Convolutional Neural Networks (CNNs) for medical image diagnosis. The authors explore the performance of ViTs versus CNNs across three medical imaging datasets, including APTOS 2019, a dataset for Diabetic Retinopathy detection. They compare the models under three initialization strategies: random initialization, ImageNet pretraining, and self-supervised learning. The results show that CNNs outperform ViTs when trained from scratch, but when pretrained on ImageNet, ViTs match CNNs in performance. Additionally, ViTs show a small advantage when combined with self-supervised pretraining.

2.1.1.2 Critical Analysis

The paper provides a valuable comparison between CNNs and ViTs for medical imaging tasks, which is a timely and important investigation given the rising popularity of ViTs. A strength of the research is the inclusion of self-supervised learning and transfer learning strategies, both of which are highly relevant to domains like medical imaging where labeled data is limited. However, the study reveals that ViTs generally require large amounts of data or pretraining to perform on par with CNNs, which could be a limitation for smaller datasets. Moreover, while ViTs provide better explainability through attention maps, the improvements in performance over CNNs are marginal in most cases with low-data regime.

2.1.1.3 Relationship to the proposed research work

The findings of this paper support the use of CNNs for feature extraction in low-data regimes, as CNNs outperformed ViTs when trained from scratch. Moreover, the insights into ViTs' performance when pretrained on large datasets like ImageNet align with our strategy of using CNNs for local feature extraction and ViTs for global attention. This hybrid approach allows us to leverage the strengths of both models, combining the robust feature extraction capabilities of CNNs with the attention-based global feature capturing power of ViTs. Additionally, the paper's emphasis on the explainability of ViTs through attention maps could enhance the interpretability of our hybrid model.

2.1.2 Conv-ViT: A Convolution and Vision Transformer-Based Hybrid Feature Extraction Method for Retinal Disease Detection

2.1.2.1 Summary

The paper[2] presents a hybrid model combining CNNs and ViT to detect retinal diseases. The model integrates pre-trained CNN models (Inception-V3 and ResNet-50) to extract texture-based features and a ViT to capture shape-based features from optical coherence tomography (OCT) images. The fusion of CNN and ViT features enhances the ability to classify retinal diseases into four categories: Choroidal Neovascularization (CNV), Diabetic Macular Edema (DME), DRUSEN, and NORMAL. The model achieves a weighted average classification accuracy of 94%, indicating the effectiveness of combining both local texture and global shape features.

2.1.2.2 Critical analysis

The Conv-ViT model presents a novel approach by fusing CNN-based texture feature extraction with transformer-based shape feature learning. A key strength of the study is the use of both Inception-V3 and ResNet-50 CNN models to address texture features while adding the ViT to capture long-range pixel relationships. This hybrid approach improves the robustness of retinal disease classification, outperforming single-model approaches. However, the complexity of the model (with three deep learning models) may limit its practicality for real-time applications due to computational demands. Additionally, the dataset used is imbalanced, which may have affected the performance on minority classes like DRUSEN, which had a lower F1 score.

2.1.2.3 Relationship to the proposed research work

The Conv-ViT method approach leverages the strengths of CNNs for extracting local features and ViTs for capturing global relationships, making it relevant for medical image analysis tasks. The insights from this paper, particularly the success of combining multiple CNNs with ViTs, support our decision to employ a hybrid model for Diabetic Retinopathy detection. Additionally, its focus on fusing CNN and ViT outputs to improve classification performance can be applied to our work, potentially enhancing the accuracy of our model.

2.1.3 Scopeformer: n-CNN-ViT Hybrid Model for Intracranial

2.1.3.1 Summary

The research[3] paper proposes a hybrid architecture combining multiple CNNs with a ViT for classifying intracranial hemorrhages. The authors utilize a feature generation backbone comprising several CNNs, including Xception networks, to create feature maps that enhance the ViT's attention mechanism. The CNNs are pretrained using different paradigms such as ImageNet and GAN-based datasets, enabling the ViT to attend to rich and multi-level feature representations. This hybrid model achieves a classification accuracy of 98.04%, demonstrating its effectiveness and scalability for medical image analysis.

2.1.3.2 Critical analysis

The research highlights a novel combination of CNN and ViT models, showcasing the strength of using CNNs as feature extractors for the ViT. One of the strengths is the use of multiple pretrained CNNs, which improves the diversity and richness of the feature maps. However, while the hybrid model is scalable, the computational complexity of stacking multiple CNNs could be a drawback for systems with limited resources. Nonetheless, the paper effectively demonstrates the benefit of integrating CNNs into ViT-based architectures for complex medical tasks.

2.1.3.3 Relationship to the proposed research work

The approach of using CNNs as feature extractors to improve the ViT's attention mechanism is directly applicable. Both tasks involve medical image classification, where the combination of local feature extraction (via CNNs) and global attention (via ViT) can enhance model performance. The findings from this paper, such as the modular scalability and the impact of using multiple pretrained CNNs, can inform the design and optimization of the proposed Diabetic Retinopathy detection model.

2.1.4 Collaborative Consultation Doctors Model: Unifying CNN and ViT for COVID-19 Diagnostic

2.1.4.1 Summary

The paper[4] proposes a hybrid model combining CNNs and ViTs for diagnosing COVID-19 using chest X-ray and CT scan images. The model draws inspiration from the collaborative consultation process in medical diagnostics, where multiple experts consult to achieve more accurate diagnoses. The CNN extracts local spatial features, while the ViT captures global contextual information. The authors employ an early fusion strategy, where CNN and ViT features are combined and then passed through a residual neural network. The model is evaluated on two benchmark datasets—ChestXray and Clean-CC-CCII—achieving accuracy rates of 98.86% and 95.62%, respectively.

2.1.4.2 Critical analysis

This paper presents an innovative approach by merging the strengths of CNNs and ViTs in medical diagnostics, emphasizing the collaboration between local feature extraction and global attention mechanisms. One of the strengths of the research is the use of early feature fusion, which maximizes the complementary capabilities of both models, leading to superior accuracy on COVID-19 diagnosis. Additionally, the analogy to a collaborative consultation between doctors provides a compelling conceptual framework for combining different models. However, the computational complexity of using both CNN and ViT models in parallel might be a barrier for real-time or resource-constrained deployments in clinical settings.

2.1.4.3 Relationship to the proposed research work

Both models aim to combine CNNs for local feature extraction and ViTs for global context analysis. The early fusion strategy discussed in the paper offers valuable insights into how we can structure the fusion of CNN and ViT features in our model. Additionally, the paper's strong results on medical image classification tasks suggest that a similar approach could improve the performance of our hybrid model for Diabetic Retinopathy detection, especially when dealing with the complexities of retinal image classification.

2.1.5 Combining the Transformer and Convolution for Effective Brain Tumor Classification Using MRI Images

2.1.5.1 Summary

The research[5] explores a hybrid model combining CNNs and ViTs for brain tumor classification using MRI images. The authors propose a Transformer-Enhanced Convolutional Neural Network (TECNN) model, where the CNN is responsible for extracting local features, while the ViT captures global dependencies using self-attention. The paper demonstrates the effectiveness of this hybrid model on two datasets: BraTS 2018 and Figshare, achieving accuracy rates of 96.75% and 99.10%, respectively. The authors introduce Feature Fusion Module (FFM) and Intelligent Merge Module (IMM) to combine CNN feature maps with Transformer patches, which ensures that both local and global information are retained and used for classification.

2.1.5.2 Critical analysis

This research presents a strong case for hybrid architectures that leverage both CNNs and ViTs for medical image classification tasks. One of its strengths is the introduction of FFM and IMM, which addresses the semantic gap between CNN feature maps and Transformer embeddings, resulting in more effective feature fusion. The TECNN model outperforms state-of-the-art methods in the datasets it was tested on, demonstrating its robustness. However, while the model achieves high accuracy, it relies on a complex architecture that may be computationally expensive, making it challenging to implement in resource-constrained environments.

2.1.5.3 Relationship to the proposed research work

The TECNN model's approach of using CNNs for local feature extraction and ViTs for global feature extraction involves medical image classification, and the hybrid architecture's ability to combine spatial and global information can be applied to our Diabetic Retinopathy dataset. The FFM and IMM modules in TECNN offer a valuable reference for effectively merging CNN and ViT outputs, which can be integrated into our model to ensure that both local (retinal features) and global (contextual) information are preserved for accurate diagnosis. This paper's methodology and findings can provide insights into optimizing our hybrid model for enhanced performance.

2.2 Analysis Summary of research items

Paper	Summary of Critical Analysis
Is it Time to Replace CNNs with Transformers for Medical Images?	This paper highlights that CNNs generally outperform ViTs in medical imaging tasks with limited data, though ViTs can match CNN performance when pretrained on large datasets like ImageNet. A strength of the study is the inclusion of self-supervised learning, which is valuable for medical images. However, ViTs require significant pretraining and computational resources, making them less practical for smaller datasets or constrained environments. While ViTs offer better interpretability with attention maps, their performance gains over CNNs are marginal in low-data scenarios
Conv-ViT: A Convolution and Vision Transformer-Based Hybrid Feature Extraction Method for Retinal Disease Detection	This paper combines CNNs and ViTs for retinal disease detection, leveraging CNNs for texture features and ViTs for shape features. This hybrid approach improves classification but increases computational complexity, which could limit real-world applications. The model also struggles with imbalanced datasets, particularly underperforming on minority classes like DRUSEN.
Scopeformer: n-CNN-ViT Hybrid Model for Intracranial	This paper introduces a hybrid model combining CNNs and ViTs for intracranial hemorrhage classification. The model leverages multiple CNNs for detailed local feature extraction and a ViT for capturing global relationships in CT images. While this approach improves classification accuracy, the reliance on multiple pretrained CNNs and the ViT increases computational complexity, which could limit scalability.

Continued on next page

Paper	Summary of Critical Analysis
Collaborative Consultation Doctors Model: Unifying CNN and ViT for COVID-19 Diagnostic	The paper presents a hybrid CNN-ViT model for COVID-19 diagnosis from medical images, combining CNNs for local feature extraction and ViTs for global attention. This collaborative approach mimics how doctors consult to improve diagnosis accuracy. While the model demonstrates strong performance in COVID-19 detection, the combination of multiple models increases computational complexity, which may hinder real-time application.
Combining the Transformer and Convolution for Effective Brain Tumor Classification Using MRI Images	This paper proposes a hybrid model integrating CNNs for local feature extraction and ViTs for capturing global context in MRI images. This combination improves classification accuracy for brain tumors by leveraging the strengths of both architectures. However, the model's complexity and high computational demands could limit its deployment in real-time or resource-constrained environments.

Table 2.1: Summary of Critical Analyses of Papers

Chapter 3

Proposed Method

This section outlines our method for Diabetic Retinopathy detection using a hybrid model that combines CNNs for local feature extraction and ViTs for global attention-based processing.

3.1 Overview

The proposed method involves the following main steps:

1. Input Image Data
2. Preprocessing and Data Augmentation
3. CNN Feature Extraction
4. ViT Attention Encoding
5. Classification and Prediction

A diagram of the approach is shown in Figure 3.1 for clarity.

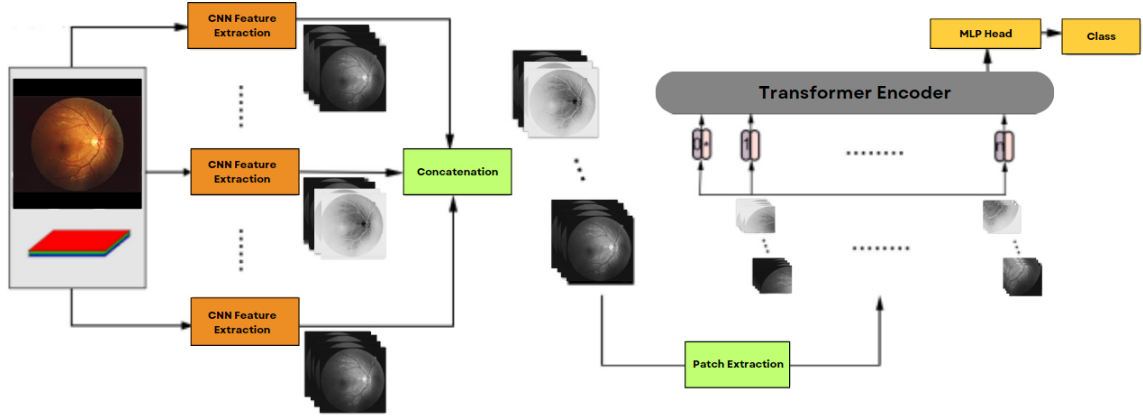


Figure 3.1: Framework of the Proposed Multi-Modal Deepfake Detection System

3.2 Methodology

Our methodology for Diabetic Retinopathy detection involves a hybrid deep learning model that integrates the strengths of CNNs and ViTs. This method leverages CNNs for local feature extraction and ViTs for capturing global dependencies in retinal fundus images, enhancing classification accuracy for Diabetic Retinopathy stages.

3.2.1 Input Image Data

We begin by using retinal fundus images as input to the model. The images are resized to a standard resolution of $224 \times 224 \times 3$ to maintain uniformity across the dataset and facilitate efficient processing by both the CNN and ViT components.

3.2.2 Preprocessing and Data Augmentation

Prior to analysis, the images undergo preprocessing to improve the quality and robustness of the model:

- **Normalization:** Pixel values are normalized to reduce intensity variations across images.
- **Data Augmentation:** Techniques such as random rotation, flipping, and zooming are applied to enhance the model's ability to generalize and reduce overfitting.

3.2.3 CNN Feature Extraction

To extract local features from the retinal images, we employ multiple pre-trained CNN models (e.g., ResNet-50, DenseNet-201) that capture fine-grained details such as blood vessel patterns,

exudates and microaneurysms. The feature maps generated by each CNN are combined and aligned to provide a comprehensive representation of the retinal images.

3.2.4 ViT Attention Encoding

After the CNN feature extraction step, the resulting feature maps are passed into a Vision Transformer (ViT) model. The ViT processes these feature maps through self-attention layers to capture long-range dependencies and global contextual information, ensuring that the model can analyze the entire image holistically.

3.2.5 Classification and Prediction

Finally, the output from the ViT is passed through a Multi-Layer Perceptron (MLP) for classification. The MLP is responsible for predicting the presence and severity of Diabetic Retinopathy based on the combined local and global features extracted by the CNN and ViT components. The model outputs the predicted class labels corresponding to different stages of Diabetic Retinopathy.

This hybrid CNN-ViT model is designed to leverage the strengths of both architectures to achieve more accurate and robust Diabetic Retinopathy detection and classification.

Chapter 4

Conclusions and Future Work

This project proposes a hybrid model that combines Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to enhance the detection and classification of Diabetic Retinopathy. The proposed approach aims to leverage CNNs for local feature extraction and ViTs for capturing global dependencies, addressing the limitations of each architecture individually. While the model has not been fully developed, the design demonstrates the potential for improving diagnostic accuracy in early detection of Diabetic Retinopathy. By integrating both deep learning techniques, this approach could provide a more efficient and accurate tool for healthcare professionals in the future. Further research and development are required to validate this approach through real-world testing and optimization for practical deployment.

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